

Worksheet 2: The Stem & The Leaf

NCERT Class 11 Biology — Chapter 5: Morphology of Flowering Plants

Student Name: _____	Roll No.: _____	Section: _____
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Section A: Multiple Choice Questions (Select the single best answer)

- An axillary bud is located in the axil of:** [____]
 - Leaflets of a pinnately compound leaf
 - Petiole of both simple and compound leaves
 - Rachis and leaflets of a palmately compound leaf
 - Terminal stipules only
- Which of the following plants exhibits alternate phyllotaxy?** [____]
 - Calotropis and Guava
 - China rose, Mustard, and Sunflower
 - Alstonia and Nerium
 - Guava and Alstonia
- A swollen leaf base found in leguminous plants is referred to as:** [____]
 - Stipule
 - Pulvinus
 - Sheathing base
 - Rachis
- Palmately compound leaves where leaflets are attached at the tip of the petiole are seen in:** [____]
 - Neem
 - Silk Cotton
 - Mango
 - Guava

Section B: Fill in the Blanks & Classifications

- Dicotyledonous leaves typically possess _____ venation, while monocotyledonous leaves display _____ venation.
- The three main morphological parts of a typical leaf are the _____ (Hypopodium), _____ (Mesopodium), and _____ (Epipodium).
- When more than two leaves arise at a single node forming a whorl, the arrangement is called _____ phyllotaxy (e.g., Alstonia).

Section C: Match the Phyllotaxy & Leaf Patterns

Arrangement Pattern	Description / Definition	NCERT Standard Example
Alternate Phyllotaxy	Single leaf arises at each node alternately	_____ —
Opposite Phyllotaxy	A pair of leaves arises at each node opposite	_____ —
Whorled Phyllotaxy	More than two leaves arise at a single node	_____ —

Pinnately Compound	Leaflets attached along a common rachis	_____ —
Palmately Compound	Leaflets attached at common point (petiole tip)	_____ —

Section D: Short Answer & Analytical Questions

8. How can you morphologically distinguish an incised simple leaf from a compound leaf?

9. Explain the biological significance of a flexible petiole and a prominent midrib in leaf physiology.

Lecturer Signature: _____